

Scientific Revolution Reflection Questions

17. Thomas Kuhn's book "The Structure of Scientific Revolutions" describes how science does not always progress smoothly but instead undergoes sudden "paradigm shifts." Using the evidence from this chapter, evaluate whether the transition from classical to modern physics fits Kuhn's model of a scientific revolution. What was the "paradigm" of classical physics? What anomalies accumulated? What was the new paradigm?
 18. Evaluate the claim: "The failure of the ether theory was the most important null result in scientific history." Do you agree? Compare it to other famous null results you know of (e.g., the absence of epicycles in Kepler's solar system model, the absence of caloric in thermodynamics). What makes a null result scientifically important?
 19. Several of the revolutionary ideas in this chapter (photons, quantum jumps, the constancy of the speed of light) were initially rejected or regarded as mathematically convenient fictions by their originators or peers. What does this tell us about how scientists evaluate new ideas? What criteria eventually convinced the physics community to accept each of these ideas?
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